
Geometric Consistency as a Signal for Identifying AI-Generated Videos

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Abstract

Generative video models can produce highly realistic footage, yet they often contain subtle geometric or temporal inconsistencies. Fully reconstructing scenes to identify inconsistencies with 3D methods like NeRF is computationally expensive and unreliable for in-the-wild video. To address this, we introduce a technique that extracts geometric feature maps using Colmap’s correspondence search. Using these feature maps, we find that a 3D Convolutional Neural Network shows some ability to distinguish real videos from synthetic ones.

1 Introduction

1.1 Importance

Generative video models such as stable video diffusion [2] can produce impressive and realistic results. However, they sometimes introduce geometric inconsistencies that make scenes look unrealistic or distorted [3]. These inconsistencies can be a problem for tasks that require accurate and stable visuals, such as creative work or simulations. Detecting these inconsistencies is important for improving the quality, reliability, and detection of deepfakes.

1.2 Key Challenges

Detecting geometric inconsistencies is challenging because they are often scene dependent and vary between generative models [3, 5]. The increasing ability of generative AI models like Sora to produce highly consistent scenes [9] makes detecting these inconsistencies even more difficult. Furthermore, the limited availability of high-quality large-scale datasets reduces the robustness of the model, and the high computational cost of training 3D ConvNets presents an additional barrier to developing effective geometric and temporal representations.

1.3 Beneficiaries

This work benefits three primary groups. For researchers, our framework offers a diagnostic tool for analyzing and minimizing geometric distortions in generative video models, advancing robustness and visual fidelity. For media and content platforms, it provides an automated mechanism to authenticate video integrity and identify AI-generated material. For the broader public, it strengthens trust in digital media by enabling reliable detection of synthetic content.

2 Related Work

2.1 Chirui Chang et al. (2024) – What Matters in Detecting AI-Generated Videos like Sora?

Relation to our approach: Chang et al. [3] decompose video into three 2.5D cues — appearance (DINOv2 features), motion (RAFT optical flow), and geometry (monocular depth) — and feed these

into 3D ConvNets for real/fake classification. Our approach is inspired by their use of 3D ConvNets, but rather than relying only on 2D+time cues, we plan to train on geometric feature maps (inspired via COLMAP corresponding search), which may capture geometric inconsistencies more directly.

Strengths and weaknesses: The approach has the advantage of being relatively simple, relying on a 3D ConvNet with clear interpretability through Grad-CAM visualizations. However, it treats geometry only as monocular depth maps rather than full 3D structures, and it focuses solely on classifying videos as real or fake without measuring the degree of inconsistency.

Our project directly addresses these weaknesses by introducing geometric feature maps and using them as input to a 3D convolution classifier, enabling more precise detection of geometric inconsistencies beyond depth maps.

Evaluation: The method was evaluated both in-domain (SVD vs. real Pexels videos) and cross-domain (Sora, Pika, and Runway vs. real Pixabay). Performance was measured using binary classification accuracy and Grad-CAM visualizations. Results were strong, with over 90% accuracy in-domain and 70–80% accuracy cross-domain.

2.2 Xuanyi Li et al. (2024) – Sora Generates Videos with Stunning Geometrical Consistency

Relation to our approach: Li et al. [9] explicitly split the videos into frames and reconstruct 3D geometry from them using COLMAP (SfM) and Gaussian Splatting. Geometric consistency was then measured by reprojecting the reconstructed 3D scene into every video frame and comparing the rendered image (from the 3D Gaussian model) against the original input frame. They used metrics such as Root Mean Squared Error (RMSE) and mean re-projection error, averaged over all frames in a sequence, to quantify geometric inconsistency. Our approach builds on this idea: instead of only using these 3D reconstructions for evaluation, we will train a 3D ConvNet directly on geometric feature maps to automatically learn inconsistency patterns.

Strengths and Weaknesses: The approach is strong in that it defines interpretable geometric metrics such as RMSE, reprojection error, and point retention, and it makes effective use of established tools like COLMAP and Gaussian Splatting to build realistic 3D representations. However, the RMSE metric alone cannot incorporate other important features such as texture authenticity and motion adherence. We plan to address this by inputting the extracted geometric feature maps into a 3D Convolution Network.

Evaluation: The method was evaluated on Sora-generated videos, Pika, and Gen2 using RMSE and reprojection error metrics. The results were convincing for Sora, highlighting Sora’s improved geometric consistency compared to earlier models.

2.3 Other Historical approaches

Before the recent dominance of diffusion models like Sora and Veo, many GAN models evaluated output with the Inception Score (IS) [1], later evolving to the Frechet Video Distance (FVD) [13]. These approaches compare the distribution of features between a set of synthetic output and real data extracted through an intermediary classifier model. IS and FVD were most commonly used to evaluate model output rather than detect synthetic images or videos. Instead of a score like IS or FVD, our approach will use a classifier to classify specific videos.

3 Method

3.1 Motivation

Our approach focuses on detecting geometric inconsistencies in synthetic video by training a 3D convolutional classifier on extracted geometric features. While conventional CNNs process individual frames, 3D ConvNets [7] incorporate a temporal dimension, allowing them to model both spatial and temporal dependencies. Inspired by Chang et al. [3], who demonstrated the effectiveness of feature-based 3D ConvNets for AI-generated video detection, we design our network to operate on

structured geometric features derived from frame correspondences. Additionally, building on Li et al. [9], who employed COLMAP [12, 11] and Gaussian Splatting [8] for 3D scene reconstruction, we extract point-based representations (inliers and outliers) to capture spatial coherence across frames. Our current progress includes implementing feature extraction pipelines based on RANSAC and SIFT correspondences, along with preliminary model training using a 3D ConvNet classifier on these geometric tensors.

3.2 Pre-Processing and Feature Extraction

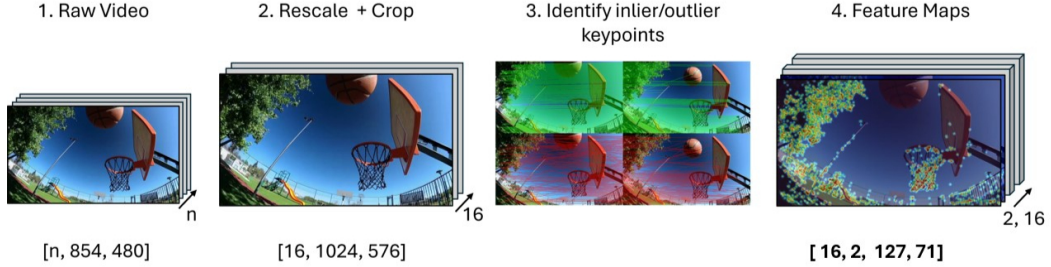


Figure 1: Pre-processing and Feature Extraction Pipeline

Our model requires video input with consistent frame count and pixel dimensions. We rescale all videos to 1024x576, applying a center-crop to those without 16:9 aspect ratios. We then take the first 16 frames at 8 FPS.

Next, we generate geometric feature maps. For each reference frame I_t in input, we select a set of neighboring frames $I_{t \pm \Delta}$, where $\Delta \in \{1, 3, 5, 10, 20\}$. If forward neighbors are unavailable near the end of a sequence, backward neighbors are used instead. The source frame and the neighbor frame are compared following the correspondence search method in COLMAP [12, 11]. Shared keypoints are identified using SIFT [10], keypoints are paired using euclidian distance brute-force matching, and finally RANSAC[6] is used to filter geometrically inconsistent keypoint matches.

Inspired by Li et al. [9], our hypothesis is that the measure of geometric consistency implied by the density of inliers and outliers identified by RANSAC[6] can be used by a deep learning model to discriminate real and synthetic video. To extract this information, we compute two-channel RANSAC[6]-based heatmaps encoding inlier and outlier densities for each frame pairing.

For each reference frame I_t and its neighboring frames $I_{t \pm \Delta}$, we detect keypoints and descriptors using SIFT[10], match them using a brute-force L2 matcher, and estimate the fundamental matrix F with RANSAC[6] to robustly separate *inliers* (geometrically consistent correspondences) from *outliers* (geometrically inconsistent matches). Each reference keypoint is assigned an *inlier score* proportional to its match quality and the frequency with which it is identified as an inlier across all neighboring frames. Similarly, an *outlier score* is computed from the ratio of RANSAC[6] rejections to the total number of neighboring frames. These values are spatially accumulated on the image plane to form two dense heatmaps representing the inlier and outlier distributions. A Gaussian smoothing kernel is then applied to reduce noise, sparsity and obtain spatially coherent maps.

Finally, each frame is represented by a two-channel tensor $[I_t, O_t]$, where:

- I_t is the inlier density map, and
- O_t is the outlier density map.

The resulting tensor $\in \mathbb{R}^{2 \times H \times W}$ captures framewise geometric consistency information that can be directly consumed by the geometry encoder of our model.

3.3 Model Details

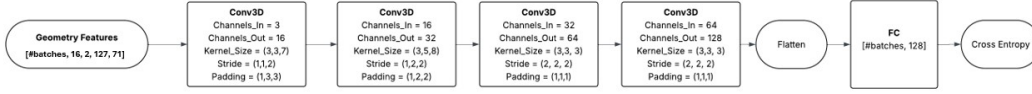


Figure 2: Model Architecture

We take as input the geometry feature maps described in section 3.2. The input passes through five 3DCNN layers with ReLU activation functions, is flattened, and finally passes through a fully-connected layer with a scalar output. Loss is calculated with binary cross entropy:

$$L = -[y \log(p) + (1 - y) \log(1 - p)] \quad (1)$$

Layer details and input shapes are outlined in Fig. 2. We train with batch size 5, learning rate $1e-5$, and weight decay $1e-5$. Additionally, to control overfitting, group normalization and dropout with $p=0.3$ is included after every CNN layer.

3.4 Obstacles

Video pre-processing and training on hundreds of GB of video data is computationally intensive, which can complicate training and limit scalability. These factors make implementing a robust and generalizable detection framework challenging. Additionally, the preprocessing step is lossy. A network trained solely on this designer input may not be able to fully capture the underlying distribution, and we may need to augment it with additional modalities, such as Appearance and Motion features, following Chang et al. [3].

4 Experiments

4.1 Datasets, Splits and Metrics

We use the GenVideo [4] dataset which supplies a training set and test set of real and synthetic video. The test set includes samples from different generators to test the generalization of the model. Due to computational constraints, we use a subset of approximately 20,000 training samples (10,000 real, 9886 fake) and 16,000 test samples (7,665 real, 8000 fake).

Train Set						
Video Source	Type	Resolution	FPS	Length	Count	Subset Used
Youku-mPLUG	Real	varies	varies	10–120s	953,279	10,000
ZeroScope	Fake	1024×576	8	3s	133,169	0
I2VGen-XL	Fake	1280×720	8	2s	61,975	0
SVD	Fake	1024×576	8	4s	149,026	9,886
VideoCrafter	Fake	1024×576	8	2s	39,485	0
Pika	Fake	1088×640	24	3s	98,377	0
DynamiCrafter	Fake	1024×576	8	3s	46,205	0
SD	Fake	512–1024	8	2–6s	200,720	0
SEINE	Fake	1024×576	8	2–4s	24,737	0
Latte	Fake	512×512	8	2s	149,979	0
OpenSora	Fake	512×512	8	2s	177,410	0

Test Set						
Video Source	Type	Resolution	FPS	Length	Count	Subset Used
MSR-VTT	Real	varies	varies	10–30s	10,000	7665
ModelScope	Fake	256×256	8	4s	700	700
MorphStudio	Fake	1280×720	8	2s	700	700
MoonValley	Fake	1024×576	16	3s	626	626
HotShot	Fake	672×384	8	1s	700	696
Show_1	Fake	576×320	8	4s	700	697
Gen2	Fake	896×512	24	4s	1,380	1380
Crafter	Fake	256×256	8	4s	1,400	1398
Lavie	Fake	1280×2048	8	2s	1,400	1396
Sora	Fake	varies	varies	60s	56	55
WildScrape	Fake	512–1024	8–16	2–6s	926	906

Table 1: GenVideo[4] dataset contents. Count is the total number of samples, and Subset Used is the number of samples used in our training run. Note: we downloaded the 100k version of the dataset, which is why the training set only contains fake samples from SVD.

4.2 Results and Baselines

Model	Avg Recall	Avg F1	Avg Accuracy
STIL	0.8222	0.7823	0.8712
VideoMAE-B	0.8750	0.8955	0.9454
MINTIME-CLIP-B	0.8762	0.8231	0.9155
FTCN-CLIP-B	0.8618	0.8755	0.9221
TALL	0.8852	0.7419	0.8791
XCLIP-B-FT	0.8441	0.8677	0.8667
DeMamba-XCLIP-FT	0.9258	0.9258	0.9710
Ours	0.7429	0.7788	0.7829

Table 2: Recall, F1, and accuracy scores for models evaluated against across all synthetic sources in the GenVideo[4] validation dataset. Sourced from the GenVideo paper[4].

4.3 Ablation and Parameter Tuning

We tested three different models to identify the architecture that can best discriminate between real and fake video utilizing our geometry feature map. All models contained the sequence of four 3D CNNs followed by a fully-connected layer. Our experiments focus on manipulating kernel parameters to grow or shrink the dimensions of sequential CNN layers.

In the Expansion model (Fig 2), the 3D CNN layers grow over time, allowing the model to reconstruct information given the geometry feature maps. In the Contraction model, the CNN layers shrink in size, forcing the model to learn increasingly higher order features. The Bottleneck model is similar to Encoder-Decoder architecture, first growing then shrinking the CNN sizes.

Model	Recall	F1	Accuracy
Expansion	0.7429	0.7788	0.7829
Contraction	0.6149	0.6701	0.6908
Bottleneck	0.7198	0.7560	0.7628

Table 3: Recall, F1, and accuracy scores for models on the GenVideo benchmark dataset leveraging our geometric feature maps.

5 Discussion and Conclusions

An accuracy of 0.78 shows that our geometry feature maps and 3D CNN model together have some capacity to discriminate between real and fake video. Although this accuracy is still below the DeMamba-XCLIP-FT baseline of 0.97 accuracy, these initial results are encouraging given a limited 20k-sample training run versus the full 1-million GenVideo training samples.

To understand the maximum capability of our proposed method, we must pre-process and train on the full 1-million-size dataset. Due to computation constants, this was unfortunately outside of the scope of this project.

We expect that our geometry feature maps alone cannot achieve SOTA performance. The method is very lossy, eliminating information such as texture and motion from the data. The original inspiration for our geometry feature maps come from feature-fusion models[3] that consider features such as appearance and motion in addition to geometry. For next steps, we propose to evaluate our geometry feature map as a component in a feature-fusion model that incorporates appearance (DINoV2) and motion (RAFT) and demonstrate accuracy increase against baselines that use other geometry signals such as monocular depth.

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